TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Practical Application of Direct, Acid, Basic and Vat Dyes

Four fibres, four binding forces, four procedures — including full vatting, dyeing and oxidation of a vat dye on cotton.

TE-3106 · TEXTILE CHEMISTRY & COLORATION-II SESSIONAL · EXPERIMENT 2 · LEVEL 3 TERM 1

1. Aim and theory

Aim: to dye four fibre–dye systems side by side and observe how application conditions follow from the binding force in each case (Theory Topic 2): **direct dye on cotton** (H-bonds + van der Waals, salt-promoted), **acid dye on wool** (ionic, acid-controlled), **basic dye on acrylic** (ionic to anionic fibre sites, retarder-controlled) and **vat dye on cotton** (reduction to soluble leuco, oxidation to entrapped pigment).

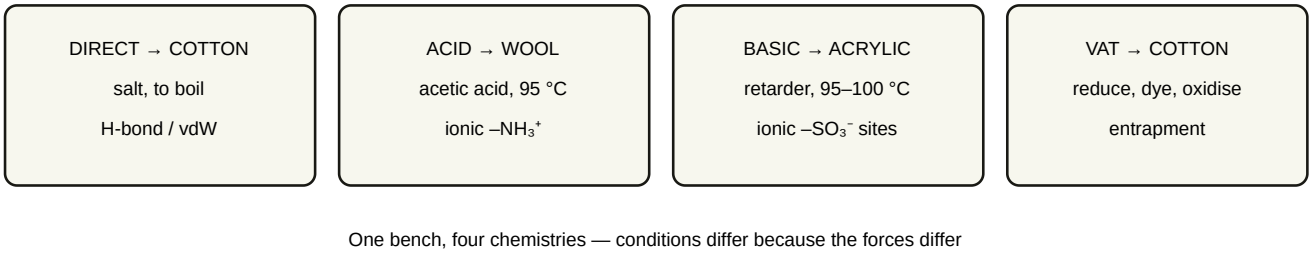


Fig. 1 — The four dye–fibre systems of this experiment.

2. Common conditions

All lots 5.0 g, LR 1:30 (bath 150 mL), dye 2.0% owf (0.100 g, dispensed from 2% stock: 5.0 mL — Experiment 1 method). Fabrics pre-wetted in hot water with 1 g/L wetting agent so every bath starts from the same state.

3. Procedure A — direct dye on cotton

1. Bath: dye stock + NaCl 15 g/L (2.25 g in 150 mL) + soda ash 2 g/L (0.30 g). Enter cotton at 40 °C.
2. Raise 1.5 °C/min to 95 °C, hold 45 min (salt may be added in two portions to avoid strike).

3. Rinse warm, then cold; dry. Note the rinse colour — unfixed direct dye leaving is the wash-fastness warning of this class.

4. Procedure B — acid dye on wool

1. Bath: dye stock + acetic acid to pH 4.5–5 (about 3% owf) + Glauber's salt 10% owf (0.50 g) as levelling agent. Enter wool at 40 °C.
2. Raise slowly (1 °C/min — wool strikes fast) to 95 °C, hold 45 min. The bath should nearly clear: anionic dye onto protonated $–NH_3^+$ sites.
3. Cool slowly to 70 °C before handling (hot wet wool felts and creases), rinse, dry flat.

5. Procedure C — basic dye on acrylic

1. Bath: dye stock + acetic acid pH 4.5–5 + cationic retarder 2% owf (0.10 g). Enter acrylic at 60 °C.
2. Raise very slowly through the strike zone 80–95 °C (0.5–1 °C/min), hold 98–100 °C 30 min. Above acrylic's wet Tg the dye rushes in; the retarder competes for the fibre's $–SO_3^-$/$–COO^-$ sites and releases them gradually.
3. Cool slowly (rapid cooling creases acrylic permanently), rinse, dry.

6. Procedure D — vat dye on cotton (vatting, dyeing, oxidation)

1. **Vatting:** paste 0.100 g vat dye with a little water; add NaOH 38 °Bé 6 mL/L (0.9 mL) and sodium hydrosulphite 5 g/L (0.75 g) in the 150 mL bath at 50–60 °C. In 10–15 min the insoluble blue pigment converts to the soluble **leuco** form (colour change to the leuco shade — e.g. indigo-type blue dye vats to a yellow-green leuco). The bath must stay reducing: sprinkle extra hydrosulphite if the leuco colour fades.

$$\text{Vat dye (insoluble quinone)} + Na_2S_2O_4 \,/\, \text{NaOH} \rightarrow \text{leuco dye (soluble, substantive)} \rightarrow [\text{air} / H_2O_2] \rightarrow \text{vat dye re-formed inside fibre}$$

2. **Dyeing:** enter cotton, hold 50–60 °C 30 min with a little salt if the recipe directs.
3. **Oxidation:** lift, expose to air 10 min (or treat with dilute H_2O_2 1–2 mL/L), watching the original dye colour return as the pigment re-forms inside the fibre pores.
4. **Soaping:** hot soap at the boil 10 min — develops the true shade and crystal form, removes surface pigment; rinse, dry.

7. Results (as recorded)

SYSTEM	BATH AT END	OBSERVATIONS
Direct/cotton	Still coloured	Good depth; rinse coloured → wash fastness the weak point, as theory predicts
Acid/wool	Almost clear	Level, full exhaustion; levelness owed to slow ramp + Glauber's salt
Basic/acrylic	Clear	Brilliant shade; a fast ramp trial piece (kept for comparison) is visibly unlevel
Vat/cotton	Leuco colour	Shade deepened through oxidation and soaping; rub-off of unsoaped piece shows why soaping is mandatory

8. Precautions

- Hydrosulphite: add to water, never water to a heap of powder; it decomposes in air — weigh last, use at once, keep the vat covered.
- Wool: no sudden temperature shocks, no boiling agitation — felting is irreversible.
- Acrylic: respect the slow ramp through 80–95 °C; it is the entire levelness story.

9. Practice (viva preparation)

1. Why is salt used in the direct bath but Glauber's salt described as a leveller in the acid bath?
2. State the chemical change in vatting and in oxidation, naming both reagents.
3. What does the retarder do, molecularly, in basic dyeing of acrylic?
4. The vat-dyed cotton is pale before oxidation and deep after soaping rather than straight after oxidation. Explain both steps.

Answer key

1. In the direct bath NaCl screens cellulose's negative charge so anionic dye can approach — it promotes exhaustion. In acid dyeing, Glauber's salt anions compete with dye anions for wool's $–NH_3^+$ sites, slowing strike — a levelling, retarding role. Same family of salt, opposite jobs, set by the fibre's charge.
2. Vatting: reduction of the insoluble quinone dye to its soluble leuco (hydroquinone) form by sodium hydrosulphite in NaOH. Oxidation: air or H_2O_2 re-oxidises leuco dye to the insoluble quinone inside the fibre.
3. The cationic retarder occupies the fibre's anionic sites first, then exchanges slowly with dye cations as temperature rises — metering dye access so uptake stays level through the strike zone.

4. Oxidation only re-forms the pigment; its crystals are initially fine and the shade dull. Hot soaping ripens crystal growth (Ostwald ripening) to the stable form with full depth and brilliance, and strips surface pigment.
